

Spectral Triples on the Sierpinski Gasket

Fabio Cipriani

Dipartimento di Matematica
Politecnico di Milano - Italy

(Joint works with *D. Guido, T. Isola, J.-L. Sauvageot*)

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Sierpinski gasket $K \subset \mathbb{C}$ and its Dirichlet form $(\mathcal{E}, \mathcal{F})$



Duomo di Amalfi: Chiostro, sec. XIII

The aim of the work is to construct Spectral Triples $(\mathcal{A}, D, \mathcal{H})$ able to describe the geometric features of K such as:

Dimension, Topology, Volume, Distance and Energy.

Our constructions rely on two facts:

- the Dirichlet space $\mathcal{F} \subset C(K)$, domain of the Dirichlet form $(\mathcal{E}, \mathcal{F})$ of the standard Laplacian on K , is a semisimple Banach $*$ -algebra so that its K -theory coincides with the K theory of $C(K)$ and with the K -theory of K [F.C. Pacific J. (2006)]
- the traces of finite energy functions $a \in \mathcal{F}$ on K to boundary of lacunas $\ell_\sigma \subset K$ belong to fractional Sobolev spaces $H^\alpha(\ell_\sigma)$ [A. Jonsson Math. Z. (2005)]

Novelty with respect to [Christensen-Ivan-Schrohe arXiv: 1002.3081 (2010)] is that

- the Dirichlet form can be recovered

$$\mathcal{E}[a] = \text{const. Trace}_\omega (|[D, a]|^2 |D|^{-\delta_D}) \quad a \in \mathcal{F}$$

- a new *energy dimension* $\delta_D := 2 - \frac{\ln 5/3}{\ln 2}$ appears as the abscissa of convergence of the *energy functionals*

$$\mathbb{C} \ni s \mapsto \text{Trace}, (|[D, a]|^2 |D|^{-s}) \quad a \in \mathcal{F}$$

The difference w.r.t. the Hausdorff dimension

$$\delta_D \neq d_H = \frac{\ln 3}{\ln 2}$$

is a consequence of the equivalent facts

- Volume and Energy are distributed singularly on K [M. Hino Prob Th. Rel. Fields (2005)]
- there are no nontrivial algebras in the domain of the Laplacian

Spectral Triples (Connes 1985)

The notion of Spectral Triple abstracts some basic properties of a Dirac operator on a complete Riemannian manifold and lies at the foundation of a metric geometry in generalized settings, where the *algebra of coordinates of the space* maybe commutative or not.

A Spectral Triple $(\mathcal{A}, \mathcal{H}, D)$ over a C^* -algebra $A \subseteq \mathbb{B}(\mathcal{H})$, consists of a *smooth* sub- $*$ -algebra $\mathcal{A} \subset A$ and a self-adjoint operator $(D, \text{dom}(D))$ on $\mathcal{H}$ subject to:

- $[D, a]$ is a *bounded operator* for all $a \in \mathcal{A}$
- $a(I + D^2)^{-1/2}$ is a *compact operator* on $\mathcal{H}$ for all $a \in A$

In the framework of the Sierpinski gasket K

- $A := C(K)$ is the *commutative* C^* -algebra of continuous functions
- $\mathcal{A} := \mathcal{F}$ is the domain of the Dirichlet form $(\mathcal{E}, \mathcal{F})$ of the standard Laplacian.

Parity, Summability and Dimensional Spectrum of Spectral Triples

- A Spectral Triple is called *even* if there exists $\gamma \in \mathbb{B}(\mathcal{H})$ such that

$$\gamma^2 = I, \quad \gamma = \gamma^*, \quad \gamma D + D\gamma = 0, \quad \gamma a = a\gamma, \quad \forall a \in A$$

- a Spectral Triple is called *finitely p-summable*, for some $p \geq 1$, if

$$a(I + D^2)^{-1/2} \in \mathcal{L}^p(\mathcal{H})$$

- a Spectral Triple is called (d, ∞) -*summable* if

$$a(I + D^2)^{-d/2} \in \mathcal{L}^{(1, \infty)}(\mathcal{H})$$

- a p-summable Spectral Triple has *discrete spectrum* if there exists a discrete set $F \subset \mathbb{C}$ such that all *zeta functions* associated to elements $a \in A$

$$\zeta_D^a : \{z \in \mathbb{C} : \operatorname{Re} z \geq p\} \rightarrow \mathbb{C} \quad \zeta_D^a(s) := \operatorname{Tr}(a|D|^{-s})$$

extends to meromorphic functions with poles contained in F . The *dimensional spectrum* of $(\mathcal{A}, \mathcal{H}, D)$ is defined to be the smallest of such sets.

Example: Canonical Spectral Triple of a Riemannian manifold M

- $\mathcal{H} := L^2(\Lambda(M))$ be the Hilbert space of square integrable sections of the exterior bundle of differential forms considered as a module over $A := C_0(M)$
- $D := d + d^*$ is the first order differential (*Dirac*) operator whose square is the Hodge-de Rham Laplacian $\Delta_{HdR} = dd^* + d^*d$ of M
- $a(I + D^2)^{-1/2}$ is a compact by Sobolev embeddings for all $a \in \mathcal{A} := C_c^\infty(M)$
- norm of commutators coincides with Lipschitz semi-norms $\|[D, a]\| = \|da\|_\infty$
- the Riemannian distance may be recovered from commutators as

$$d(x, y) := \sup\{a(x) - a(y) : a \in \mathcal{A}, \|[D, a]\| \leq 1\}$$

- the Minakshisundaram-Pleijel asymptotic of the heat semigroup $e^{-t\Delta_{HdR}}$ as $t \rightarrow 0$ allows to identify the dimensional spectrum with

$$\{0, 1, \dots, \dim M\} \subset \mathbb{C}$$

so that the spectral triple is $(\dim M, \infty)$ -summable.

Fredholm modules (Atiyah (1969), Kasparov (1975))

Fredholm modules on C^* -algebras generalize elliptic operators on manifolds:

A Fredholm module $(F, \mathcal{H})$ over a C^* -algebra $A \subseteq \mathbb{B}(\mathcal{H})$, consists of a

- *symmetry* $F \in \mathbb{B}(\mathcal{H})$: $F^* = F$, $F^2 = I$ such that
- $[F, a]$ is compact for all $a \in A$
- FM lie at the core of Noncommutative Geometry of A. Connes where the compact operator $da := i[F, a]$ is the *operator theoretic substitute* for the differential of $a \in A$
- FM has been constructed on p.c.f. fractals by [C.-Sauvageot CMP (2009)] and on more general fractals by [Ionescu-Rogers-Teplyaev (2011)]

Proposition. (Connes (1986), Baaj-Julg (1983))

The sign $F := \text{sign}(D)$ of the Dirac operator of a Spectral Triple $(\mathcal{A}, \mathcal{H}, D)$ gives rise to a Fredholm module $(F, \mathcal{H})$.

Pairing with K-theory and an Index Theorem on SG

Modifying the proof of the Connes Index Theorem for FM [Connes (1986)], to take into account that Dirac operators on SG have

$$\dim \operatorname{Ker} (D) = +\infty ,$$

it is possible to construct topological invariants.

Theorem. (CGIS 2010)

The Spectral Triples on SG determine a group homomorphism on the K-theory

$$\operatorname{ch}_D : K_1(\mathcal{F}) \rightarrow \mathbb{C}$$

which is integer valued $\operatorname{ch}_D(K_1(\mathcal{F})) \subset \mathbb{Z}$.

These values are the index of suitable Fredholm operators:

- *for all invertible $u \in GL_\infty(\mathcal{F})$, setting $P := (I + F)/2$, the operator $G_u := PuP$ is Fredholm and*

$$\operatorname{ch}_D[u] = \operatorname{Index} (G_u) = \frac{1}{2^{2m+1}} \operatorname{Trace} (u[F, u^{-1}]^{2m+1}) \quad \forall m \geq 1 .$$

These values can be interpreted as winding numbers around lacunas in SG.

Quasi-circles

We will need to consider on the 1-torus $\mathbb{T} = \{z \in \mathbb{C} : |z| = 1\}$ structures of quasi-circle associated to the following Dirichlet forms and their associated Spectral Triples for any $\alpha \in (0, 1)$.

Lemma. *Fractional Dirichlet forms on a circle (CGIS 2010)*

Consider the Dirichlet form on $L^2(\mathbb{T})$ defined on the fractional Sobolev space

$$\mathcal{E}_\alpha[a] := \int_{\mathbb{T}} \int_{\mathbb{T}} \varphi_\alpha(z-w) |a(z) - a(w)|^2 dz dw \quad \mathcal{F}_\alpha := \{a \in L^2(\mathbb{T}) : \mathcal{E}_\alpha[a] < +\infty\}$$

where φ_α is the Clausen cosine function for $0 < \alpha \leq 1$.

Then $\mathcal{H}_\alpha := L^2(\mathbb{T} \times \mathbb{T})$ is a symmetric Hilbert $C(K)$ -bimodule w.r.t. actions and involutions given by

$$(a\xi)(z, w) := a(z)\xi(z, w), \quad (\xi a)(z, w) := \xi(z, w)a(w), \quad (\mathcal{J}\xi)(z, w) := \overline{\xi(w, z)}.$$

The derivation $\partial_\alpha : \mathcal{F}_\alpha \rightarrow \mathcal{H}_\alpha$ associated to $\mathcal{E}_\alpha$ is given by

$$\partial_\alpha(a)(z, w) := \varphi_\alpha(z - w)(a(z) - a(w)).$$

Proposition. *Spectral Triples on a circle (CGIS 2010)*

Consider on the Hilbert space $\mathcal{K}_\alpha := L^2(\mathbb{T} \times \mathbb{T}) \oplus L^2(\mathbb{T})$, the left $C(\mathbb{T})$ -module structure resulting from the sum of those of $L^2(\mathbb{T} \times \mathbb{T})$ and $L^2(\mathbb{T})$ and the operator

$$D_\alpha := \begin{pmatrix} 0 & \partial_\alpha \\ \partial_\alpha^* & 0 \end{pmatrix}.$$

Then $\mathcal{A}_\alpha := \{a \in C(\mathbb{T}) : \sup_{z \in \mathbb{T}} \int_{\mathbb{T}} \frac{|a(z) - a(w)|^2}{|z - w|^{2\alpha + 1}} < +\infty\}$ is a uniformly dense subalgebra of $C(\mathbb{T})$ and $(\mathcal{A}_\alpha, D_\alpha, \mathcal{K}_\alpha)$ is a densely defined Spectral Triple on $C(\mathbb{T})$. The dimensional spectrum is $\{\frac{1}{\alpha}\}$.

Dirac operators on K .

Identifying isometrically the main lacuna $\ell_\emptyset$ of the gasket with the circle $\mathbb{T}$, consider the Dirac operator $(C(K), D_\emptyset, \mathcal{K}_\emptyset)$ where

- $\mathcal{K}_\emptyset := L^2(\ell_\emptyset \times \ell_\emptyset) \oplus L^2(\ell_\emptyset)$
- $D_\emptyset := D_\alpha$
- the action of $C(K)$ is given by restriction $\pi_\emptyset(a)b := a|_{\ell_\emptyset}$.

Fix $\gamma > 0$ and for $\sigma \in \Sigma$ consider the Dirac operators $(C(K), \pi_\sigma, D_\sigma, \mathcal{K}_\sigma)$ where

- $\mathcal{K}_\sigma := \mathcal{K}_\emptyset$
- $D_\sigma := 2^{\gamma|\sigma|} D_\alpha$
- the action of $C(K)$ is given by contraction/restriction $\pi_\sigma(a)b := (a \circ F_\sigma)|_{\ell_\emptyset} b$.

Finally, consider the Dirac operator $(C(K), \pi, D, \mathcal{K})$ where

- $\mathcal{K} := \bigoplus_{\sigma \in \Sigma} \mathcal{K}_\sigma$
- $\pi := \bigoplus_{\sigma \in \Sigma} \pi_\sigma$
- $D := \bigoplus_{\sigma \in \Sigma} D_\sigma$

Notice that $\dim \text{Ker } D = +\infty$ and that D^{-1} will be defined to be zero on $\text{Ker } D$.

Volume functionals and their Spectral dimensions

Theorem. (CGIS 2010)

The zeta function $\mathcal{Z}_D$ of the Dirac operator $(C(K), D, \mathcal{K})$, i.e. the meromorphic extension of the function $\mathbb{C} \ni s \mapsto \text{Trace}(|D|^{-s})$ is given by

$$\mathcal{Z}_D(s) = \frac{4}{1 - 3c^{-s}} z(\alpha s)$$

where z denotes the Riemann zeta function. The dimensional spectrum is given by

$$\mathcal{S}_{dim} = \left\{ \frac{1}{\alpha} \right\} \cup \left\{ \frac{\ln 3}{\ln 2} \gamma^{-1} \left(1 + \frac{2\pi i}{\ln 3} k \right) : k \in \mathbb{Z} \right\}$$

and the abscissa of convergence is $d_D = \max(\alpha^{-1}, \frac{\ln 3}{\ln 2} \gamma^{-1})$. When $0 < \gamma < \frac{\ln 3}{\ln 2} \alpha$ there is a simple pole in $d_D = \frac{\ln 3}{\ln 2} \gamma^{-1}$ and the residue of the meromorphic extension of $\mathbb{C} \ni s \mapsto \text{Trace}(f|D|^{-s})$ gives the Hausdorff measure of dimension $d_H = \frac{\ln 3}{\ln 2}$

$$\text{Trace}_{Dix}(f|D|^{-s}) = \text{Res}_{s=d_D} \text{Trace}(f|D|^{-s}) = \frac{4d}{\ln 3} \frac{z(d)}{(2\pi)^d} \int_K f d\mu.$$

Notice the complex dimensions and the independence of the residue Hausdorff measure upon $\gamma > 0$.

Spectral Triples and Connes metrics on the Sierpinski gasket

Theorem. (CGIS 2010)

$(C(K), D, \mathcal{K})$ is a Spectral Triple for any $0 < \gamma \leq 1$. In particular the seminorm

$$a \mapsto \|[D, a]\|$$

is a Lip-seminorm in the sense of Rieffel so that the distance it determines induces the original topology on K .

If $\alpha < \gamma$ this distance is bi-Lipschitz w.r.t. the power $(\rho_{\text{geo}})^\gamma$.

Energy functionals and their Spectral dimensions

By the Spectral Triple it is possible to recover, in addition to dimension, volume measure and metric, also the energy form of K

Theorem. (CGIS 2010)

Consider the Spectral Triple $(C(K), D, \mathcal{K})$ for $\alpha \leq \alpha_0 := \frac{\ln 5}{\ln 4} - \frac{1}{2} \left(\frac{\ln 3}{\ln 2} - 1 \right) \sim 0,87$ and assume $a \in \mathcal{F}$. Then the abscissa of convergence of

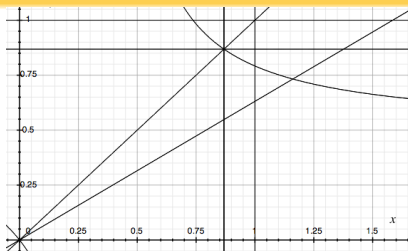
$$\mathbb{C} \ni s \mapsto \text{Trace}(|[D, a]|^2 |D|^{-s})$$

is $\delta_D := \max(\alpha^{-1}, 2 - \gamma^{-1} \frac{\ln 5/3}{\ln 2})$.

- If $\delta_D > \alpha^{-1}$ then $s = \delta_D$ is a simple pole and the residue is proportional to the Dirichlet form

$$\text{Res}_{s=\delta_D} \text{Trace}(|[D, a]|^2 |D|^{-s}) = \text{const. } \mathcal{E}[a] \quad a \in \mathcal{F};$$

The (γ, α) -plane



- Above the line $\alpha = \gamma/d_S$, the volume measure is a multiple of the Hausdorff measure H_{d_S} where $d_S = \frac{\ln 3}{\ln 2}$ and the spectral dimension is $d_D = \frac{d_S}{\gamma}$
- in the square $0 < \alpha, \gamma \leq 1$, $(\mathcal{A}, D, \mathcal{H})$ is a Spectral Triple whose Fredholm module has nontrivial pairing with all generators of the K-theory $\mathbb{K}^1(K)$ and "generates" the K-homology $\mathbb{K}^0(K)$
- below the line $\alpha = \gamma$, the distance ρ_D is bi-Lipschitz with $(\rho_{\text{geo}})^\gamma$
- in the strip $\alpha < \alpha_0 < \gamma < 1$ and above the hyperbola $\alpha \delta_D(\gamma) = 1$ the energy functional is proportional to the Dirichlet form, where $\delta_D(\gamma) = 2 - \frac{\ln(5/3)}{\ln 2 \ln} \gamma^{-1}$ is the energy dimension.